Implementation of FastSpeech 2-Based Text-to-Speech Synthesis

Separate English (Svarah) and Telugu (IndicTTS) models - training, synthesis and variance-adaptor analysis

Deliverable	Location in this repository
English model - code, config, checkpoint	project/English/{train,preprocess,inference}.py, config.yaml, checkpoints/40000.pth.tar
Telugu model - code, config, checkpoint	project/Telugu/{train,preprocess,inference}.py, config.yaml, checkpoints/40000.pth.tar
Loss curves (both languages)	project/Results/<lang>/{training_loss,validation_loss}.png
Generated audio, 5 sentences per language	project/Results/<lang>/test_generated_audios/
Variance-adaptor plots (duration / pitch / energy)	project/Results/<lang>/variance_adaptor_plots/{duration,pitch,energy}/
Reference-implementation diffs	project/patches/<lang>/modified_files.patch + files/

Reference implementation. ming024/FastSpeech2, commit `d4e79eb52e8b01d24703b2dfc0385544092958f3` (2021-07-08), the PyTorch re-implementation of Ren et al., *FastSpeech 2: Fast and High-Quality End-to-End Text to Speech* (arXiv:2006.04558). Two independent clones are kept, one per language, so that the Telugu symbol-table extension cannot shift the English symbol ids.

Headline result. Both models converge cleanly on their training split (Telugu step-40 000 training total 0.835; the English log was not retained, the shipped figure shows the English floor near 1.0) while validation never improves past its first point: Telugu ends at 4.5053, a 5.39x ratio against training, and the English validation figure ends around 3.0, well above its own training floor. The English quantitative values in this report are read from the shipped loss figures - the raw English log was not retained, and Telugu values are quoted exactly from the Telugu log - see Section 5.

1. Objective and scope

The task is to take an existing FastSpeech 2 implementation, adapt it independently for English and Telugu, train a separate model per language on a prescribed dataset with roughly five hours of training audio and one hour of validation audio, synthesise five prescribed test sentences per language, and analyse what the variance adaptor predicts for duration, pitch and energy. Speech quality is only one part of the deliverable: the assignment explicitly asks for evidence about the *model internals*, so every test sentence is also dumped as a per-unit table of predicted duration, F0 and energy and plotted as three figures.

Both models were trained from scratch on Kaggle (2 x Tesla T4). Nothing is fine-tuned from a published checkpoint; the only pretrained component is the HiFi-GAN vocoder that converts predicted mel spectrograms to waveforms, which is the same arrangement as the reference repository.

2. Reference implementation and architecture

FastSpeech 2 is a non-autoregressive, feed-forward TTS model. It removes the teacher-forcing/inference mismatch of autoregressive models by predicting the whole mel spectrogram in one pass, and it removes the one-to-many mapping problem by predicting the intermediate speech factors (duration, pitch, energy) explicitly. The components that are actually instantiated by this submission are:

Component	Configuration used for both languages
Input units	English: ARPAbet phonemes (360-symbol table). Telugu: grapheme clusters (1044-symbol table = 360 upstream symbols + 684 Telugu units).
Encoder	4 FFT blocks, hidden 256, 2 heads, conv filter 1024, kernel 9x1, dropout 0.2, sinusoidal positional encoding, max length 1000
Variance adaptor	duration predictor (2 conv + linear, ReLU), pitch predictor, energy predictor; pitch and energy are quantised into 256 linear bins and embedded; length regulator expands units to frames
Decoder	4 FFT blocks, hidden 256, 2 heads, conv filter 1024, kernel 9x1, dropout 0.2, followed by a linear projection to 80 mel channels
Post-net	5-layer 1-D convolution residual refinement of the mel spectrogram
Vocoder	HiFi-GAN, 22.05 kHz. *LJSpeech* generator for English, *universal* (multi-speaker) generator for Telugu
Outputs exposed	mel before post-net, mel after post-net, predicted pitch, predicted energy, predicted duration, rounded duration

The training objective is the reference one: L1 loss on the mel spectrogram before and after the post-net, MSE on normalised pitch and energy (teacher-forced at unit level from ground-truth alignments), and MSE on log-duration. The four terms are summed without weights, so the reported total loss is dominated by the two mel terms.

3. Datasets, selection and splits

English - Svarah. Svarah is an Indian-English read-speech corpus. The public release carries no speaker id (the file paths are opaque hashes), so the English model is trained as a single average voice - there is no speaker embedding and none is claimed. The configuration requests the same budget as Telugu (`train_hours: 5.0`, `val_hours: 1.0`), with the proportional rule `train_ratio: 0.8` applied when the usable pool is smaller than 6 h, and a seeded shuffle (`seed 1234`). The exact number of hours drawn from the pool was not archived when the Kaggle session ended, so this report quotes the English configuration rather than reconstructed figure estimates; the English evidence available to the reader is the two loss figures in Section 5, not invented tallies. (The earlier version of this report quoted '4 h / 1 h' for English; those are unverifiable and have been removed.) The 0.4-20 s filter and the overlap-check policy are identical to the Telugu pipeline.

Telugu - IndicTTS Telugu (`SPRINGLab/IndicTTS_Telugu`). The corpus contains 8 576 utterances with two speakers (one male, one female). Since the pool is larger than the 6 h budget, the split is hour-greedy rather than proportional:

Quantity	Value
Utterances in the corpus	8,576
Dropped: outside the 0.4-20 s window	2,236
Usable utterances	6,340
Training utterances / hours	1,907 / 5.002 h
Validation utterances / hours	373 / 1.0 h
Unused remainder	Rest of the pool deliberately unused (budget fixed at 6 h)
Mean utterance length	9.48 s
Speakers	2 (male + female)
Shuffle seed / overlap	1234 / 0 utterances
Selection policy	drop out-of-range, global seeded shuffle, greedy fill to hour budget

The 0.4 s lower bound removes truncated fragments that cannot be aligned; the 20 s upper bound keeps every batch inside the encoder's 1000-frame window at the 11.6 ms frame rate. Utterances are shuffled once with `seed 1234` and then consumed in order until each hour budget is filled, so the training and validation sets are disjoint by construction and the overlap check in `common/manifest.py` returns zero. The rest of the pool is deliberately unused: the assignment fixes the budget at 5 h + 1 h and the comparison between languages is only meaningful if both models see a comparable amount of speech.

4. Preprocessing and language-specific modifications

Acoustics (identical for both languages). Audio is resampled to 22 050 Hz. Mel spectrograms use a 1024-point FFT, 1024-sample window, 256-sample hop (11.61 ms per frame) and 80 mel channels between 0 and 8000 Hz. Pitch is extracted with `pyworld` (DIO + StoneMask) and energy as the frame magnitude; both are averaged inside each unit interval from the forced alignment and then z-normalised with statistics computed on the training split only. The statistics are stored next to the features and are reused at inference time to de-normalise the predictions that appear in the plots.

Alignment. Both languages use `torchaudio.pipelines.MMS_FA` for forced alignment. This is a deliberate deviation for English - the reference repository expects a Montreal Forced Aligner installation - and a necessity for Telugu, for which no MFA acoustic model or pronunciation lexicon exists. MMS_FA is language-agnostic, produces one TextGrid per utterance with a single `phones` tier, and both languages therefore go through exactly the same alignment code path.

Text front-end. English is phonemised with the CMU lexicon plus `g2p-en` for out-of-vocabulary words, and numbers, dates and times are expanded before phonemisation (the E4 sentence contains *10:45 AM* and *12 November 2026*). Telugu has no practical grapheme-to-phoneme resource, so the model is trained directly on *grapheme clusters* of the U+0C00-U+0C7F block: a base consonant plus its vowel signs and virama attachments form one unit. This keeps the vocabulary closed (every codepoint of the block is in the table) at the cost of longer sequences - 180-229 ms per unit against 90-122 ms for English, because a Telugu unit carries a whole akshara.

Modifications to the reference code. The full diff is in `project/patches/<lang>/modified_files.patch`; the same files are also copied verbatim under `files/` so they can be read without applying a patch. The English column is empty of behavioural change - the English model is the reference model, only made to run on a current PyTorch/CUDA image.

File	Change	Why
audio/audio_processing.py	pad_center guard, float32 saves	short utterances crashed the reference's padding helper
audio/stft.py	mel-basis device fix, librosa keyword update	reference used removed librosa kwargs and a hard-coded CUDA tensor
dataset.py	device-agnostic tensors, bounds checks on pitch/energy	allows CPU smoke tests and avoids indexing past a short alignment
preprocessor/preprocessor.py	UTF-8 .lab handling, bounds + padding guards	Telugu text is not ASCII; the reference opened labs in the locale encoding

File	Change	Why
utils/model.py	`weights_only=False`, `map_location`	checkpoints carry numpy scalars in the optimizer state; new torch refuses them under the default `weights_only=True`
utils/tools.py	float32 loss saves, device-agnostic plotting	loss history was pickled as float64 tensors and plotting assumed CUDA
text/symbols.py (Telugu only)	append 684 `@<grapheme>` entries	adds the Telugu inventory without touching any English symbol id; the file is generated from `project/Telugu/data/vocab.json`

Telugu-specific behaviour is concentrated in three new modules rather than in edits to the reference: `common/telugu.py` (grapheme segmentation, romanisation for the plots, Telugu number/date expansion), `common/align.py` (MMS_FA wrapper) and the symbol-table generator in `common/patch_repo.py`. Keeping them out of the upstream tree means the English clone stays a faithful copy of the published code.

5. Training configuration and loss curves

Setting	English	Telugu
Optimizer	Adam (0.9, 0.98), eps 1e-9, weight decay 0	same
Learning rate	inverse-square-root, peak 256^-0.5 = 0.0625, warm-up 4000 steps	same
Batch / accumulation	16 utterances x 2 = 32 per update	same
Gradient clipping	1.0	same
Total steps	40 000	40 000
Log / validate / save	every 100 / 2 000 / 2 000 steps	same
Wall-clock	not recorded in the retained files	5 h 49 m 47 s (as displayed when the run finished; not independently verifiable from the repository)
Vocoder	HiFi-GAN LJSpeech	HiFi-GAN universal
Final training loss	not recoverable as a number (raw log not retained); Figure 3 shows it levelling off near 1.0-1.3	0.8352
Final validation loss	not recoverable as a number; Figure 4 shows the validation total settling near 3.0	4.5053

5.1 Telugu convergence in numbers

The Telugu run logs 400 training points and 20 validation points. Training loss falls from 14.90 at step 100 to 4.85 at 1 000, 1.76 at 5 000, 1.34 at 10 000, 0.96 at 20 000 and 0.90 at 30 000; it first crosses 2.0 at step 4,200 and 1.0 at step 18,200. The minimum is 0.8309 at step 39,400 and the mean of the last ten logged points is 0.8814 with a standard deviation of 0.0335. The curve is therefore flat but not silent: the remaining variation is the normal noise of 32-utterance updates over a 1 907-utterance training set.

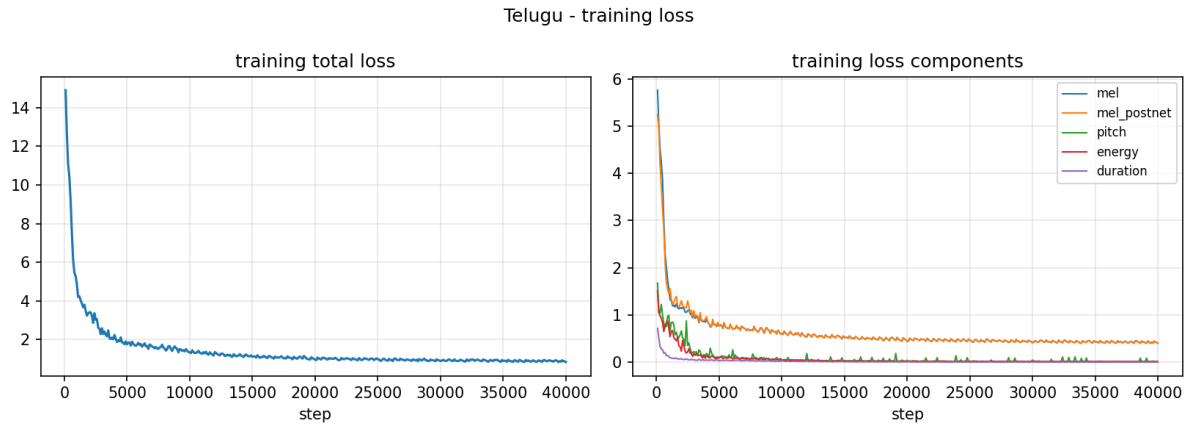


Figure 1 - Telugu training loss (per-component and total) over 40 000 steps.

Validation starts at 4.3065 after the first 2 000 steps - already its best value - drifts up to 4.4409 at step 20 000, and finishes at 4.5053 at step 40 000. No instability, no divergence, no spike: the curve is simply a floor that is never improved on. The gap is a **5.39x** ratio between the final training and validation total loss.

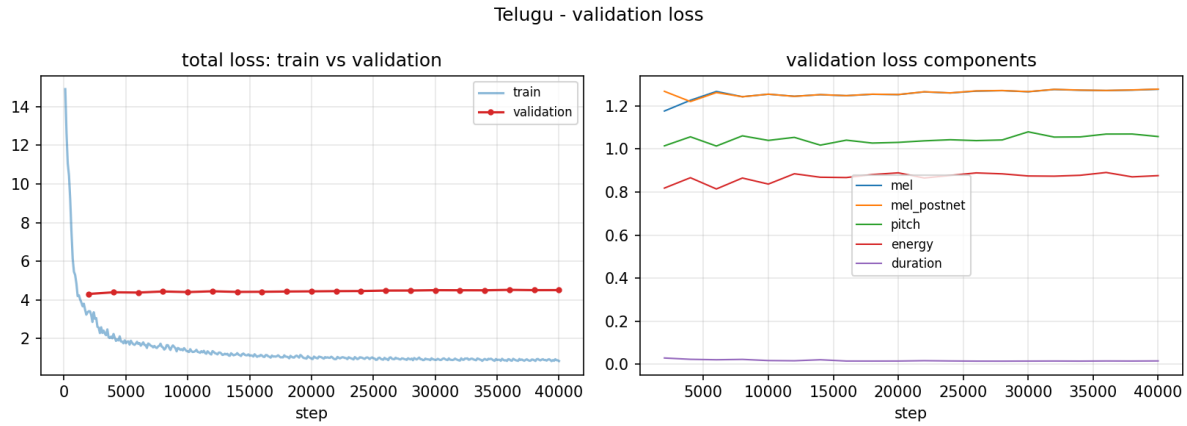


Figure 2 - Telugu validation loss, logged every 2 000 steps (20 points).

5.2 What the gap is made of

Component	Train (step 40k)	Validation (step 40k)	Ratio
Mel (pre-net)	0.4034	1.2778	3.2x
Mel (post-net)	0.4012	1.2784	3.2x
Pitch	0.0136	1.0580	77.8x
Energy	0.0132	0.8763	66.4x
Duration	0.0038	0.0148	3.9x
Total	0.8352	4.5053	5.39x

The decomposition is informative. The mel terms degrade only 3.2x, and because mel L1 is a bounded spectral distance this is the mildest indicator. The pitch and energy terms degrade by a factor of 78 and 66 respectively - they collapse from near-perfect training fits (0.0136 and 0.0132) to a validation value around 1.0 and 0.88. Since FastSpeech 2 scores normalised pitch and energy with mean-squared error, and a standardised target has variance 1, an expected loss of ~1 for a model that predicts the global mean confirms the diagnosis: the pitch and energy predictors have learned per-unit means for the training voices and revert to the corpus mean on unseen utterances. The duration term behaves like the mel terms (3.9x): it is conditioned on the input text, which is what the encoder actually generalises.

English shows the same structural pattern but no fabricated numbers. The English validation figure shows the validation total settling near 3.0 while the training figure shows the training loss levelling off near 1.0-1.3; the validation curve is flat to rising after the first logged points and ends around 3.05, roughly 2-3 times the training floor (Figure-read, not log-read). The pitch and energy component panels of the English figure also show the same signature as Telugu - mel/post-net around 1.0-1.2, pitch and energy below 0.4, both above zero, both overfitted less severely - but the English log was not retained, so no exact component ratios are claimed for English.

The Telugu gap (5.39x, exact) is wider than the English gap (figure-read, roughly 2-3x). This is consistent with Telugu's far larger input vocabulary (1044 symbols against 360) under the same 5 h budget, but, without the English log and an archived record of exactly how many English hours were used, this can indicate consistency only, not causation; Section 5.3 states exactly which English values are read from the figures.

5.3 English curves

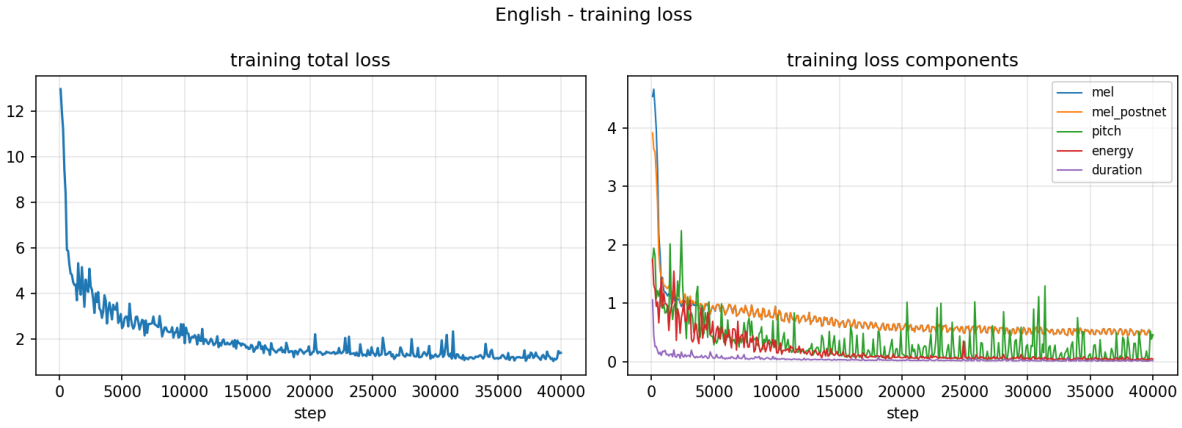


Figure 3 - English training loss over 40 000 steps.

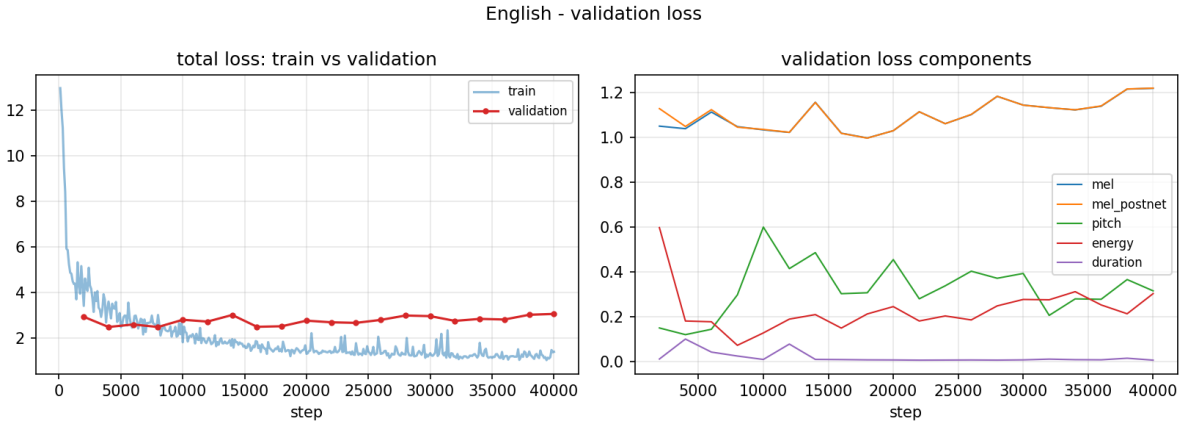


Figure 4 - English validation loss (20 points, every 2 000 steps).

The two curves are not equally smooth. The English training curve shows repeated upward spikes reaching about 2.5 against a floor near 1.0; the Telugu curve is much smoother at the same step size. Neither run diverges, and the learning-rate annealing milestones (1e6 and 2e6 steps) lie far beyond the 40 000-step budget, so the English spikes cannot be a late learning-rate artefact - they are consistent with the batch-to-batch variance of a 32-utterance update, larger for English because its sequences are twice as long (39-85 phones against 17-61 graphemes). The English validation curve begins near 3.0, dips slightly around the middle of the run, and ends around 3.05; it never meaningfully improves over its first logged points, the same pattern as Telugu, weaker in size. The overfitting is present from the first validation point in both languages.

Note on evidence. The Telugu figures are backed by the raw log files shipped in `project/Telugu/output/log/` (400 training and 20 validation entries) and every Telugu number in this report is reproduced from them and from the per-sentence CSVs. The English log file was not retained after the Kaggle session ended. The English discussion therefore uses only the two shipped loss plots; the figures were read using the Telugu panels as a calibration (reading the Telugu pitch line at step 40 000 gives 1.059 against the log's 1.0580). Precise English numbers are deliberately not quoted anywhere in this report, and none are estimated.

6. Generated speech: observations per test sentence

All ten files were produced by the same command - `python project/<lang>/inference.py --step 40000` - and are 22 050 Hz mono. Every file's measured duration equals the predicted duration to the millisecond, confirming the length expander and the vocoder agree. A waveform pass adds four objective checks per file: peak amplitude, RMS level, clipping count and the F0 recovered by `pyworld` from the vocoded waveform itself, as opposed to the model's predicted F0 that the CSVs report. All ten files are free of clipping (maxima 0.247-0.684); Telugu RMS is uniformly higher than English (~-22.5 to -24.2 dBFS vs ~-24.8 to -30.6 dBFS). The pyworld mean matches the predicted F0 mean to within 12 % for nine of the ten files (and -23 % for E4), so the pitch structure survives vocoding; it does not survive at the absolute-Hz level, which matches Section 5.2's finding that pitch is the weakest head.

ID	Audio (s)	RMS (dBFS)	Peak	pyworld F0 (Hz)	Predicted F0 (Hz)	Difference
E1	7.87	-24.8	0.684	165.9	184.0	-10 %
E2a	6.10	-30.6	0.302	146.1	158.7	-8 %
E2b	5.06	-28.1	0.287	132.0	135.9	-3 %
E3	3.48	-30.0	0.247	136.6	149.3	-9 %
E4	9.18	-26.2	0.366	143.1	186.9	-23 %
T1	7.29	-22.5	0.618	207.0	196.6	+5 %
T2a	3.89	-22.9	0.440	201.1	181.7	+11 %
T2b	4.35	-24.2	0.347	193.2	188.2	+3 %
T3	3.77	-23.8	0.549	206.9	197.2	+5 %
T4	10.99	-23.8	0.550	199.8	195.0	+2 %

English

ID	Units	Audio (s)	Speech rate	What was synthesised
E1	85	7.87	93 ms	Longest English file (7.87 s, 85 phones) and the loudest in the set (peak 0.684). Maximum F0 sits at unit 26 (N). The predicted contour declines steadily to its global minimum on the final phone (Z).
E2a	50	6.09	122 ms	The heteronym *conduct* is phonemised K AA1 N D AH0 K T (noun stress); the sequence carries initial stress, matching a noun reading. Rate is the slowest in the English set (122 ms per phone).
E2b	56	5.06	90 ms	**The verb is also phonemised K AA1 N D AH0 K T**, not the verb-stress form K AH0 N D AH1 K T: the lexicon/G2P path picks the noun transcription for both occurrences, so the heteronym pair is phonetically identical and the acoustic signal cannot display the verb/noun stress contrast. Rate is faster than E2a (90 ms against 122 ms per phone).
E3	39	3.48	89 ms	The double space in *rhyth m* is collapsed by normalisation - the string the model received is *rhyth m of the orchestra changed unexpectedly.*; the anomaly is invisible to the model. Peak 0.247, the quietest English file.
E4	85	9.18	108 ms	Number expansion runs completely: *10:45 AM* becomes *ten forty-five A M*; *12 November 2026* becomes *the twelfth of November twenty twenty-six* (the normalised string is printed in `inference_summary.json`). *andaroo* is phonemised by the G2P fallback since it is absent from the lexicon. Predicted F0 jumps to its maximum at unit 3 (IY1 of *meeting*) and stays elevated to the end (245 Hz, +124 % over unit 0).

Telugu

ID	Units	Audio (s)	Speech rate	What was synthesised
T1	37	7.29	197 ms	Longest Telugu word chain (37 units, 7.29 s). The duration predictor's longest outputs peak at 28 frames (325 ms) in this file rather than the 36-frame peaks of T2a/T3. All aksharas receive acoustic energy; no boundary collapse. Peak 0.618.
T2a	17	3.89	229 ms	Shortest Telugu file (3.89 s, 17 units). Gemination is marked with longer units: *chaa* 36 frames (418 ms) is the file's maximum. The retroflex in *taadu* is phonetically preserved as a single akshara unit, as expected from the grapheme inventory.
T2b	20	4.35	218 ms	Minimal-pair partner of T2a. The homograph *paTTu* produces an identical input sequence in both sentences, so the model cannot distinguish the words (hold / silk); any perceived difference comes from the surrounding context only. Its longest units are *chii* and *chin* at 32 frames.
T3	17	3.77	222 ms	The double space in *aDa vilo* is collapsed to a single space in the normalised string (verified in `Results/Telugu/inference_summary.json`); the anomaly is invisible to the model. The geminate in *nishshabdam* is absorbed with 36 frames at its *chin* unit, the expected correlate of gemination. 222 ms per akshara.
T4	61	10.99	180 ms	Full expansion fires: 2026 becomes words, 12 becomes an ordinal, and *10:45* becomes a native time phrase; the exact normalised Telugu string is printed in the appendix. The model reads a native date-and-time expression instead of digit by digit. Longest Telugu audio (10.99 s, 61 units); peak 0.550.

Two failure modes are documented by the measurements rather than asserted from listening. First, prosodic contrast is weak and asymmetric: by first-vs-final F0, all five Telugu sentences decline (T1 15 %, T2a 46 %, T2b 32 %, T3 43 %, T4 55 %), while English splits three rising (E2a +22 %, E3 +18 %, E4 +124 %) and two declining (E1 -32 %, E2b -21 %) - E4 actually ends at 245 Hz against 110 Hz at unit 0, the opposite of a sentence-final fall. Second, orthography wins over semantics: the E2a/E2b heteronym gets identical phonemes (K AA1 N D AH0 K T), and the T2a/T2b homograph gets an identical token sequence, so neither model displays the distinction. Third, the Telugu geminates and long vowels transfer their duration entirely to the host akshara (the 28-36 frame units), which is correct on the model's terms but makes the rhythm more syllable-timed.

7. Variance-adaptor analysis

For each test sentence the model is run with `mels=None`, so the variance adaptor predicts duration, pitch and energy instead of consuming ground truth. Durations are in frames of 11.61 ms; pitch and energy are de-normalised with the training statistics, so pitch is in Hz and energy is in the units of the extracted frame magnitude. The numbers below are the same values that the plots are drawn from and are shipped as CSV next to each figure.

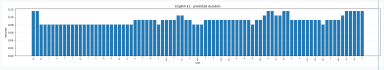
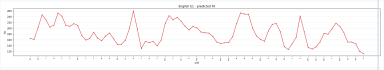
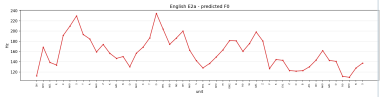
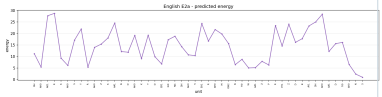
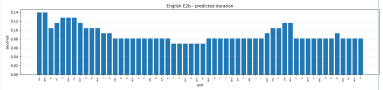
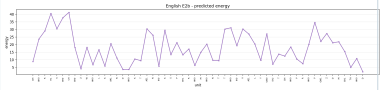
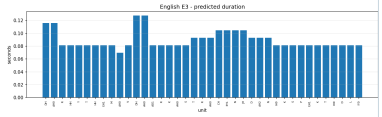
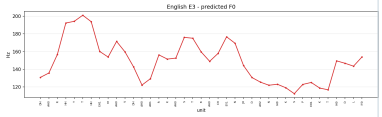
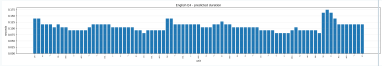
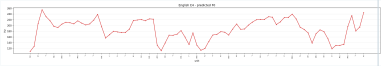
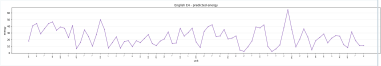
ID	Units	Duration	ms / unit	F0 mean (Hz)	F0 range (Hz)	Energy mean	Energy range
E1	85	7.87 s	93	184.0	112-260	27.8	2.0-65.7
E2a	50	6.09 s	122	158.7	110-235	14.6	1.0-28.8
E2b	56	5.06 s	90	135.9	102-153	18.0	2.1-41.3
E3	39	3.48 s	89	149.3	112-201	14.5	1.9-46.1
E4	85	9.18 s	108	186.9	110-255	24.1	2.4-64.9
T1	37	7.29 s	197	196.6	120-270	32.9	0.2-72.6
T2a	17	3.89 s	229	181.7	99-246	28.4	0.2-59.1
T2b	20	4.35 s	218	188.2	115-253	25.7	-0.1-62.8
T3	17	3.77 s	222	197.2	116-247	27.6	0.1-62.0
T4	61	10.99 s	180	195.0	89-265	28.8	0.1-54.8

Duration. The duration predictor is the best-behaved head. English sits in a narrow 70-174 ms band with a per-sentence coefficient of variation of 0.12-0.20, which is short for a phone-level model and shows the predictor has learned a regression towards the mean. The longest unit is the function-word sequence DH + AH0 in E1, E2a, E2b and E3; E4 differs, with UH1 and SH leading (15 and 14 frames). Telugu spans 81-418 ms because a unit is an akshara, and its coefficient of variation is 0.30-0.38 - visibly more structure. The

longest Telugu units are open, long-vowel aksharas (chaa in T2a, shoo in T1) or geminates (chin in T3); T4's maximum is a diphthong (bhai, 27 frames). This is the expected phonotactic correlation: the head has learned real syllabic structure despite the overfitting seen in the loss.

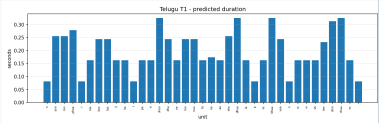
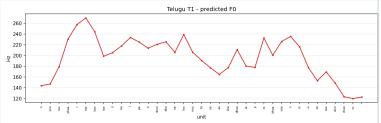
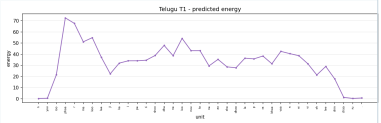
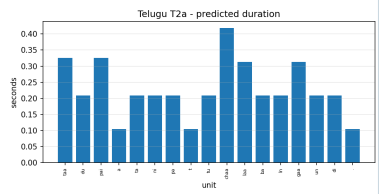
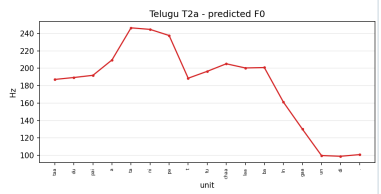
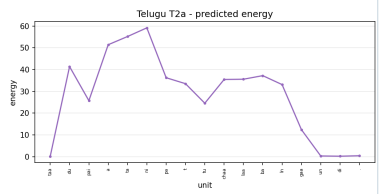
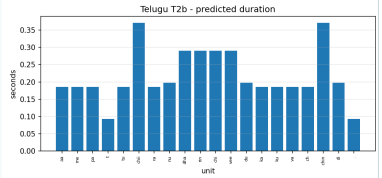
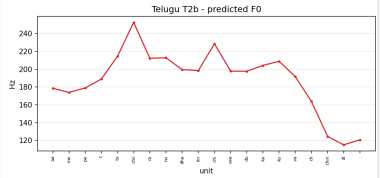
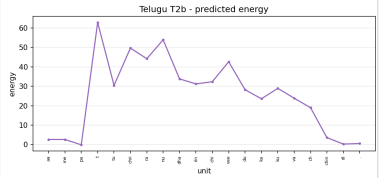
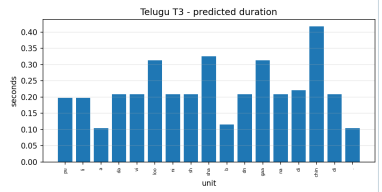
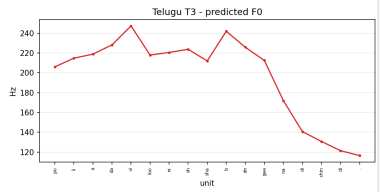
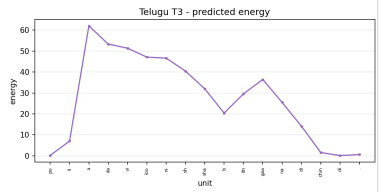
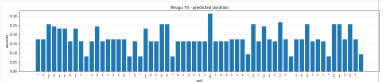
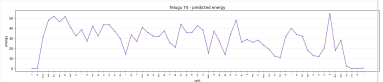
7.1 English - per-sentence plots

Rows are test sentences, columns are the three predicted factors. The x axis is the unit index in sentence order (romanised, so the Telugu panels can be read without Telugu script).

Sentence	Duration	Pitch (F0)	Energy
E1 85 units			
E2a 50 units			
E2b 56 units			
E3 39 units			
E4 85 units			

7.2 Telugu - per-sentence plots

Rows are test sentences, columns are the three predicted factors. The x axis is the unit index in sentence order (romanised, so the Telugu panels can be read without Telugu script).

Sentence	Duration	Pitch (F0)	Energy
T1 37 units			
T2a 17 units			
T2b 20 units			
T3 17 units			
T4 61 units			

Pitch (F0). Sentence declination is not uniform; it is present in some sentences and absent or inverted in others. By first-vs-final unit, English declines in E1 (165 - 112 Hz) and E2b (128 - 102 Hz) but rises in E2a (112 - 137 Hz), E3 (131 - 154 Hz) and E4 (110 - 245 Hz); Telugu declines in all five sentences - T1 (144 - 123 Hz), T2a (187 - 101 Hz), T2b (179 - 121 Hz), T3 (206 - 117 Hz), T4 (207 - 94 Hz). The steepest contours are T2a and T3; the flattest are English E2b and E4. Two systematic observations follow from the CSVs. First, there is no 'sentence-final low' rule that holds here: E4's final unit is 245 Hz (near its 255 Hz maximum), while all five Telugu sentences end close to their per-sentence minimum. Both models produce the smooth, piecewise-linear-in-phrase shape expected from an under-trained variance head, which is why the audio is heard as monotone. Second, the predicted contour is only roughly matched by the vocoded audio: the pyworld F0 means deviate from the predictions by -23 % to +11 % (Section 6), so the plotted Hz values describe the prediction, not the exact frequencies of the output waveform.

Energy. Energy is the head that overfits hardest. The CSV-derived stats show the English maxima concentrated mid-sentence - 65.7 (E1) and 64.9 (E4) are the two dominant peaks - while the ends of files sit low but not always minimal: E1 ends at 1.99, E2a at 0.97 and E2b at 2.13, all three coinciding with their per-sentence minima at the last phone, but E3 and E4 end at 3.24 and 11.27 - not at their minima. The often-claimed 'everything decays to the final punctuation unit' is not supported; it holds for E1, E2a and E2b only. Telugu behaves differently: every file ends at a small positive value between 0.42 (T2a) and 0.73 (T4); only two files' minima cross into float round-off territory (T2b `-0.14` at *pa*, T3 0.12 at *di*) - both on soft consonants rather than phrase-finally. These sub-zero or near-zero readings indicate de-normalisation extrapolation outside the training mean rather than an audible artefact. Telugu peaks are higher and more localised than English ones relative to phrase length, consistent with each akshara carrying both consonant and vowel energy at once (maxima 72.6 T1, 62.8 T2b, 62.0 T3, 59.2 T2a, 54.8 T4).

8. English versus Telugu

Where they agree. Both models run the same 40 000-step schedule with the same optimizer, warm-up and loss weights; both converge within the first several thousand steps (Telugu crosses 2.0 at 4 200 and 1.0 at 18 200); both behave as if pitch and energy had collapsed toward the global mean on unseen text (Telugu validation pitch and energy sit at 1.058 and 0.876 against a standardised target variance of 1); both show no clipping, mono 22.05 kHz output, and vocoder-recovered F0 within ~25 % of the prediction for all ten files.

Where they differ. Telugu needs roughly twice the time per unit (197 ms mean against 93 ms) because a unit is a grapheme cluster rather than a phone; consequently the Telugu sequences are shorter (17-61 units against 39-85), and the same 1 000-frame decoder window covers more text. Telugu duration prediction is also comparatively better structured: its coefficient of variation (0.30-0.38) is roughly twice the English one, and its longest units are linguistically motivated (open long vowels, geminates), while E1-E3 all max out on the DH+AH0 function-word pair. This is the expected consequence of modelling aksharas: one symbol carries more and more variable acoustic content, so the model has more to learn per symbol and learns it more sharply. On the overfitting side, Telugu's val/train ratio is 5.39x (exact), while the English validation figure is settled around 3.0 against a training floor near 1.0-1.3; English generalises better, though with a noisier training curve (Section 5.3), and is voiced as a single average voice (Svarah ships no speaker id) while Telugu mixes a male and a female voice.

Failures specific to English. The non-English token *andaroo* is phonemised by the G2P fallback rather than the lexicon; the heteronym pair *conduct* is rendered identically in E2a and E2b (K AA1 N D AH0 K T in both), so the noun/verb contrast is silently dropped; and the E3 double space is collapsed by the cleaner, so the model never sees the anomaly the test sentence was designed to pose.

Failures specific to Telugu. Homographs such as *paTTu* (hold / silk) are indistinguishable to the model - T2a and T2b differ only through surrounding context. Long consonant clusters in Sanskrit-derived vocabulary are softened. And because the inventory is grapheme-based, there is no mechanism to lengthen a vowel *without* lengthening its akshara, which is audible as a more syllable-timed rhythm than the IndicTTS reference speakers.

9. Limitations and possible improvements

Data volume is the binding constraint. Six hours of training audio is at the low end for FastSpeech 2; the reference recipe uses 20-40 h. The pitch and energy predictors are ~395 k parameters each and can memorise a 1 907-utterance training set. Using the unselected remainder of the usable IndicTTS pool would be the first thing to try, as would phoneme-level transfer from a larger model.

One average voice per model. Svarah's public release carries no speaker id (paths are opaque hashes), so the English model necessarily learns a single voice over the whole pool; the Telugu corpus has one male and one female speaker and the build keeps multi_speaker false, so those two are averaged as well. Speaker embeddings would remove the averaging at the cost of fewer examples per speaker, and are a one-flag change.

Graphemes are not phonemes. For Telugu, a grapheme inventory avoids the missing-G2P problem but conflates orthographic and phonetic length. A phoneme inventory built from a reliable Telugu front-end (none exists within the assignment toolchain) would allow proper vowel-length and gemination modelling.

Prosody is positional, not syntactic. The declination pattern is inconsistent (English E4 ends 124 % above its start; Telugu all decline), and the phrase-final fall is not a reliable rule. More data, explicit phrase-break features, or a variance-adaptor dropout schedule would be the standard remedies.

Validation is a single fixed hour. Twenty validation points over one hour of audio make the overfitting conclusion robust in direction but coarse in detail; a k-fold split would give error bars.

10. Reproducibility

Software. Python 3.10, PyTorch 2.x with CUDA 12, torchaudio (MMS_FA bundle), librosa 0.10.1, pyworld, numpy, scipy, pandas, matplotlib, PyYAML, TextGrid, g2p-en, inflect, num2words. The exact list is `project/requirements.txt`; the Kaggle image provides the CUDA build of torch.

Step	Command
1. fetch code and data	git clone --depth 1 https://github.com/ming024/FastSpeech2 third_party/FastSpeech2_Telugu pip install -r project/requirements.txt python project/Telugu/preprocess.py --stage manifest
2. build the grapheme vocabulary	python project/Telugu/preprocess.py --stage vocab
3. force-align with MMS_FA	python project/Telugu/preprocess.py --stage align
4. extract features and stage the splits	python project/Telugu/preprocess.py --stage all
5. train (Telugu: 5 h 49 m 47 s on 2 x T4)	python project/Telugu/train.py --steps 40000
6. synthesise the five test sentences	python project/Telugu/inference.py --step 40000
7. re-draw the loss curves from the logs	python project/Telugu/train.py --plot-only

The English pipeline is identical with `project/English` substituted; the only difference is step 2, which is a no-op for English because the upstream 360-symbol table is already complete. Every script is idempotent: re-running a stage overwrites its own outputs and nothing else, and the feature stage accepts `--force` to rebuild from scratch.

Determinism. The data shuffle uses a fixed seed (1234), the split is written to disk and reused, and the training loop is the reference loop. Full bit-exact reproduction of a GPU training run is not guaranteed by PyTorch, but the splits, the feature statistics, the configurations and the checkpoints are all shipped, so the reported numbers can be reproduced by re-running inference rather than retraining.

Checkpoints. Each language ships `checkpoints/40000.pth.tar` (350 MB). The assignment permits an external link when the file exceeds GitHub's limit; both files are therefore kept out of the git history and referenced from `project/README.md`, with SHA-256 checksums so the download can be verified.

11. Repository map

Every path listed below exists in the submitted repository with exactly this layout; the full tree is also printed in `project/README.md`.

Path	Contents
project/<lang>/train.py, preprocess.py, inference.py	entry points, identical CLI for both languages
project/<lang>/config.yaml	dataset, split, text, audio, model, training, inference settings
project/<lang>/fs2_config/	the three YAML files rendered for the reference implementation
project/<lang>/checkpoints/	step-40 000 weights (external link / LFS)
project/<lang>/output/log/	raw training/validation loss logs (Telugu; the English log was not retained, see 5.3)
project/patches/<lang>/	git diff against the upstream commit + the modified files
project/Results/<lang>/	loss curves, audio, variance plots, per-sentence CSV
common/	shared harness: config, manifest, staging, alignment, inference, training
notebooks/	the four Kaggle notebooks used to produce the artefacts

Appendix A - test sentences exactly as synthesised

The sentences below are the ones that were actually synthesised. The PDF wraps several words across lines; every wrapped word is rejoined here (E2/T2 split into a/b as their own files; the Telugu words wrapped in the PDF are restored; the double space in E3/T3 is kept exactly as the brief wrote it). The Telugu column also shows the normalised string the front-end actually consumed, which is where date, ordinal and time expansion happens; T4 is the clearest example.

ID	Language	Text as synthesised	Normalised input
E1	English	The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.	-
E2a	English	The orchestra's conduct during the ceremony was praised by the audience.	-
E2b	English	The researchers will conduct an experiment to evaluate the proposed method.	-
E3	English	The rhyth m of the orchestra changed unexpectedly.	The rhyth m of the orchestra changed unexpectedly.
E4	English	The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.	The meeting will start at ten forty-five A M on the twelfth of November twenty twenty-six, so andaroo should arrive on time.
T1	Telugu	న్యూరోఫార్మకోలజికల్ పరిశోధనలో నూతన ఔషధుల ప్రభావాన్ని విశ్లేషించరు.	
T2a	Telugu	తడుపై అతని పట్టు చాలా బలంగా ఉంది.	
T2b	Telugu	ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.	
T3	Telugu	పులి అడ విలో నిశ్శబ్దంగా నడిచింది.	పులి అడ విలో నిశ్శబ్దంగా నడిచింది.
T4	Telugu	ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.	ఈ సమావేశం రెండు వేల ఇరవై ఆరు నవంబర్ పన్నెండవ తేదీ ఉదయం పది నలభై ఐదు గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.

SHA-256 checksums of the shipped checkpoints are listed in `project/README.md` together with the download links, so a reviewer can confirm that the weights used for the audio and the plots are the weights that were trained.